

MAT 534 HW03

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Problem 1. Let G be a finite group, P a Sylow p -subgroup of G . If N is a normal subgroup of G , show that $P \cap N$ is a Sylow p -subgroup of N , and PN/N is a Sylow p -subgroup of G/N .

Solution. The key fact is that if H is a finite group and $Q \leq H$ is a subgroup with order a power of p such that $(H : Q)$ isn't divisible by p , then Q is a Sylow p -subgroup of H (since its order is therefore the largest power of p that divides $|H|$).

By HW02 #1 we have

$$(N : P \cap N) = (PN : P) \mid (G : P).$$

Since the order of P is the largest power of p dividing G , the index of P in G is not a multiple of p , hence neither is $(N : P \cap N)$. Since $P \cap N$ is a subgroup of N it has prime power order, so these two facts together imply that $P \cap N$ is a Sylow p -subgroup of N .

Furthermore, $(PN : N) = (P : P \cap N)$ is a power of p , so PN/N has order a power of p , and

$$(G/N : PN/N) = (G : PN) \mid (G : P)$$

is not a multiple of P , so PN/N is a Sylow p -subgroup of G/N . $\square$

Problem 2. Let G be a finite group, p a prime, and H a p -subgroup of G . Show that there is a Sylow p -subgroup of G containing H .

Solution. This is essentially just our definition of Sylow p -subgroups: a Sylow p -subgroup is a p -subgroup of G that is maximal under inclusion, and H , by definition of maximality, must be contained in a maximal (Sylow) p -subgroup of G . $\square$

Problem 3. Let P and Q be Sylow p -subgroups of a finite group G . Suppose the center $Z(P)$ of P is contained in Q and is normal in Q . Prove that $Z(P) = Z(Q)$.

Solution. Let $H = N_G(Z(P))$ be the normalizer of $Z(P)$. Then P and Q are Sylow p -subgroups of H because the largest power of p dividing $|H|$ is at most the largest power of p dividing $|G|$, so there exists $h \in H$ such that $Q = hPh^{-1}$. At the same time, $hZ(P)h^{-1} = Z(P)$ because h is in the normalizer of $Z(P)$.

The conjugation-by- h map ϕ from P to Q (i.e. $\phi(x) = h x h^{-1}$) is an isomorphism and it maps the center of P bijectively to the center of Q : to verify this we see that $ab = ba$ for $a \in Z(P)$, $b \in P$ implies $\phi(a)\phi(b) = \phi(b)\phi(a)$ where $\phi(a) \in hZ(P)h^{-1}$ and $\phi(b) \in Q$; conversely, $ab = ba$ for $a \in hZ(P)h^{-1}$, $b \in hPh^{-1} = Q$ implies $\phi^{-1}(a)\phi^{-1}(b) = \phi^{-1}(b)\phi^{-1}(a)$ where $\phi^{-1}(a) \in Z(P)$ and $\phi^{-1}(b) \in P$. In other words, $Z(Q) = hZ(P)h^{-1} = Z(P)$ as desired. $\square$

Problem 4. Let G be a group of order p^2q , where p and q are distinct primes. Prove that G has a normal Sylow subgroup.

Solution. Let n_p be the number of Sylow p -subgroups and n_q be the number of Sylow q -subgroups, meaning

$$\begin{aligned} n_p &\equiv 1 \pmod{p}, & n_p &\mid q \\ n_q &\equiv 1 \pmod{q}, & n_q &\mid p^2. \end{aligned}$$

If either n_p or n_q are 1 then we are done, so assume otherwise. In this case $n_p = q$ and $n_q = p$ or p^2 . Thus $p \mid q - 1$ by the $1 \pmod{p}$ constraint and $n_q \equiv 1 \pmod{q}$ means that q divides either $p - 1$ or $p^2 - 1$, but $q \mid p^2 - 1$ encompasses both cases. That is, we want to find prime numbers p and q such that $p \mid q - 1$ and $q \mid p^2 - 1$. Since q divides $p^2 - 1 = (p - 1)(p + 1)$, it divides one of $p - 1$ and $p + 1$. However q cannot divide $p - 1$ because $p \mid q - 1$ means that $q > p > p - 1$, so we have $q \mid p + 1$. This in turn implies $q \leq p + 1$, which, in addition to $p \leq q - 1$, implies that $p + 1 = q$. Thus the only pair of primes satisfying the divisibility constraints is $(p, q) = (2, 3)$.

Now, let P_2 and P_3 be respectively Sylow 2- and 3-subgroups of G . Their intersection $P_2 \cap P_3$ must be the trivial group because its order must divide $\gcd(|P_2|, |P_3|) = \gcd(2, 3) = 1$. By the number-theoretic restrictions, n_2 is either 1 or 3 and n_3 is either 1 or 4. If $n_3 = 1$ we are done, so suppose $n_3 = 4$. Any two distinct Sylow 3-subgroups intersect only at the identity because Sylow 3-subgroups are cyclic, so together the four Sylow 3-subgroups contain $1 + 2 \cdot 4 = 9$ elements of G . At this point there is only room for one Sylow 2-subgroup, namely the identity element plus the three elements not in any of the Sylow 3-subgroups, since any other hypothetical Sylow 2-subgroup would intersect a Sylow 3-subgroup at a non-identity element, which we argued above is impossible. Thus either n_2 or n_3 is 1, which is what it means for either the Sylow 2-subgroup or Sylow 3-subgroup to be normal in G . $\square$

Problem 5. Let G be a finite group, P a Sylow p -subgroup of G . Suppose that H is a subgroup of G containing $N(P)$. Show that $(G : H) \equiv 1 \pmod{p}$.

Solution. We know that the normalizer of P in H is the same as the normalizer of P in G because H contains $N(P)$. Furthermore, since Sylow p -subgroups of G are all conjugate to one another, the number of Sylow p -subgroups is $(G : N(P))$, is $1 \pmod{p}$. P must also be a Sylow p -subgroup of H , so similarly $(H : N(P)) \equiv 1 \pmod{p}$. Therefore

$$(G : H) = (G : N(P)) / (H : N(P)) \equiv 1/1 = 1 \pmod{p}. \quad \square$$

Problem 6. Let K be a normal subgroup of a finite group G . If P is a Sylow p -subgroup of K , prove that $G = K \cdot N_G(P)$, where $N_G(P)$ is the normalizer of P in G .

Solution. For every $g \in G$, the conjugate gPg^{-1} of P must be a subgroup of K because of the normality of K . Then P and hence gPg^{-1} are Sylow p -subgroups of K , so they are conjugate in K , i.e. there exists $k \in K$ such that $gPg^{-1} = kPk^{-1}$. Equivalently, $k^{-1}g \in N_G(P)$ so $g \in kN_G(P)$, i.e. any $g \in G$ is contained in some coset of $N_G(P)$ by an element of K , which is what $G = K \cdot N_G(P)$ means. $\square$

Problem 7. Let G be a finite group of order 80. Show that G is not simple.

Solution. Let $H \leq G$ be a Sylow 2-subgroup, so $|H| = 16$. Then, $|G| = 80$ does not divide $(G : H)! = 120$, so HW02 #6(d) implies that G is not simple. $\square$

Problem 8. Suppose that the number n_p of Sylow p -subgroups of a finite group G satisfies $n_p \not\equiv 1 \pmod{p^2}$. Prove that there are two distinct Sylow p -subgroups P and Q such that $|P \cap Q| = |P|/p$.

Solution. Fix an arbitrary Sylow p -subgroup P and let it act by conjugation on the set X of Sylow p -subgroups. Then, the number of elements of any orbit of this action is a power of p (because it divides $|P|$). This action has one fixed point (namely P) and k orbits of size p ; all other orbits have size p^m for $m \geq 2$, so the sum of the sizes of the orbits is

$$|X| = n_p \equiv 1 + kp \pmod{p^2},$$

which is not $1 \pmod{p^2}$, so $p \nmid k$ and in particular $k \neq 0$. Pick $Q \in X$ whose orbit has size p . The stabilizer of Q then has size $|P|/p$ by orbit-stabilizer. If $pQp^{-1} = Q$ for some $p \in P$ then $p \in P \cap N_G(Q)$, but since the order of $p \in N_G(Q)$ is a power of p , by the lemma we proved in class, $p \in Q$. Therefore the stabilizer of Q is $P \cap Q$, so $|P \cap Q| = |P|/p$. $\square$

Problem 9. Let $G = N_0 \supseteq N_1 \supseteq \cdots \supseteq N_r = \{1\}$ be a **composition series** for a finite group G , meaning that each N_{i+1} is a normal subgroup of N_i , and that the factor groups N_i/N_{i+1} are all simple. Suppose that one of the N_i s is a Sylow p -subgroup of G (for some prime p). Prove that N_i must be normal in G .

Solution. Let $P = N_i$ be a Sylow p -subgroup of G ; this means P is a Sylow p -subgroup of $N_1, \dots, N_{i-1}$ as well. We prove that if P is normal in N_{i-j} for some $1 \leq j \leq i-1$, then P is also normal in $N_{i-(j+1)}$ via induction on j .

For the base case $j = 1$, we have that P is normal in N_{i-1} and N_{i-1} is normal in N_{i-2} . Consider the conjugates of P under N_{i-2} ; these are the Sylow p -subgroups of N_{i-2} . The normality of N_{i-1} means that conjugates of P stay subgroups of N_{i-1} , hence are Sylow p -subgroups; this means that N_{i-2} also has P as its only Sylow p -subgroup, hence P is normal in N_{i-2} .

The proof in the general case is similar; P is normal in N_{i-j} , so conjugation by N_{i-j-1} sends P to Sylow p -subgroups of N_{i-j} , which are still P , i.e. P is fixed under conjugation by N_{i-j-1} so it is normal in N_{i-j-1} .

Thus taking for $j = i$ tells us that P is normal in $N_0 = G$. $\square$

Problem 10. Suppose G is a nontrivial finite p -group.

- (a) Show that the center of G is nontrivial.
- (b) Show that G is simple if and only if $|G| = p$.
- (c) Let $|G| = p^n$ and let $G = N_0 \supseteq N_1 \supseteq \cdots \supseteq N_r = \{1\}$ be a composition series for G . Show that $r = n$ and $|N_i| = p^{n-i}$ for $i = 0, 1, \dots, n$.
- (d) Show that we can find a composition series for G such that each N_i is normal in G .

Solution. (a) The sizes of the conjugacy classes of G are divisors of $|G|$, hence powers of p , that sum to the order of G . The center is comprised of the fixed points under conjugation, the number of which has the same remainder mod p as $|G|$, namely zero. That is, $|Z(G)|$ is a multiple of p and is at least 1 because it contains the identity of G , so $Z(G)$ must be nontrivial.

(b) If $|G| > p$ then either $Z(G) \neq G$ and so $Z(G)$ is a nontrivial (not equal to G or $\{1\}$) normal subgroup, or $Z(G) = G$ in which case G is abelian; here any subgroup is normal and there exists a subgroup isomorphic to $\mathbb{Z}/p\mathbb{Z}$ by Cauchy's theorem (which is not equal to G because its order is too small).

On the other hand if $|G| = p$ then any subgroup has order dividing p , hence the order of the subgroup is 1 or p , which means that the subgroup is trivial or it is G . Since G has no nontrivial proper subgroups it certainly can't have nontrivial proper *normal* subgroups.

(c) The orders of the N_i are all powers of p so N_{i-1}/N_i must have order p in order to be simple due to part (b). This implies that $|N_{i-1}| = |N_i|/p$, so by induction we get $|N_i| = p^{n-i}$ for all $i = 0, 1, \dots, n$ and $r = n$ because the order reaches 1 after n divisions by p .

(d) Let $|G| = p^n$. We induct on n , with $n = 1$ trivial because $|G| = p$ means G is simple.

In the general case, the center of G is nontrivial and has order a power of p , so it has a subgroup N isomorphic to $\mathbb{Z}/p\mathbb{Z}$; this subgroup is normal in G because its elements commute with all the elements of G (due to being a subgroup of the center). Then $G/N = \overline{G}$ has order p^{n-1} , so it has a composition series of subgroups $\overline{G} \supseteq \overline{N}_1 \supseteq \dots \supseteq \overline{N}_r = \{N\}$ that are all normal in G (where $\overline{N}_r$ consists of just the one coset of N). By the fourth isomorphism theorem, the subgroups $\overline{N}_i$ in this chain all correspond to normal subgroups N_i of G that contain N , and $N_{i-1}/N_i \cong \overline{N}_{i-1}/\overline{N}_i$ is simple for all $i = 1, \dots, r$. Thus, lifting the $\overline{N}_i$ to normal subgroups of G , we get the composition series of G

$$G = N_0 \supseteq N_1 \supseteq \dots \supseteq N_r = N \supseteq \{1\}$$

with only normal subgroups of G . □